05_intro_machine_learning

March 4, 2026

1 Introduction to Machine Learning

1.1 A 90-Minute Journey for Actuaries

Course Duration: 90 minutes
Target Audience: Actuaries and quantitative professionals
Learning Objective: Understand core ML paradigms, model selection, and practical implementation

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1.1.1 What We'll Cover Today

This session introduces the fundamental concepts and practical tools for machine learning, with special emphasis on applications relevant to the actuarial profession. By the end, you will understand:

- The three learning paradigms and when to apply each
- How to select appropriate model complexity
- How to encode, scale, and engineer features
- How to build, evaluate, and optimize classification and regression models

[1]:
```
# Essential imports
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.neural_network import MLPClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler, MinMaxScaler, OneHotEncoder
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.metrics import classification_report, confusion_matrix,␣
↪accuracy_score
from sklearn.metrics import precision_score, recall_score, f1_score
import warnings
warnings.filterwarnings('ignore')

# Import helper functions from mytools
from mytools import (
```

```
    generate_polynomial_data,
    generate_two_blobs,
    plot_model_selection_regression,
    plot_classification_boundaries,
    generate_multiclass_data,
    plot_confusion_matrix_custom,
    load_titanic_data
)

# Set visualization defaults
#sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)
plt.rcParams['font.size'] = 11

# Avoid truncation of DataFrame output in Jupyter
pd.set_option('display.expand_frame_repr', False)

# Display all columns in the DataFrame output
pd.set_option('display.max_columns', None)

# Define a separator for better readability
sep = "=" * 50

print("All imports successful!")
```

```
All imports successful!
```

1.2 1. Learning Paradigms

Machine learning encompasses three primary paradigms, each suited to different types of problems:

1.2.1 Three Core Paradigms

Paradigm	Description	Use Case
Supervised Learning	Learn from labeled data: input X → output y	Predicting claims, customer churn
Unsupervised Learning	Find patterns in unlabeled data	Customer segmentation, anomaly detection
Reinforcement Learning	Learn through interaction & rewards	Optimal decision-making, control systems

In this session, we focus on **Supervised Learning**, which splits into two main branches:

- **Classification:** Predict categorical outcomes (e.g., "Will claim occur?" → Yes/No)
- **Regression:** Predict continuous outcomes (e.g., "What will be the claim amount?" → €5,000)

1.3 2. Model Selection — Regression (Underfitting vs. Overfitting)

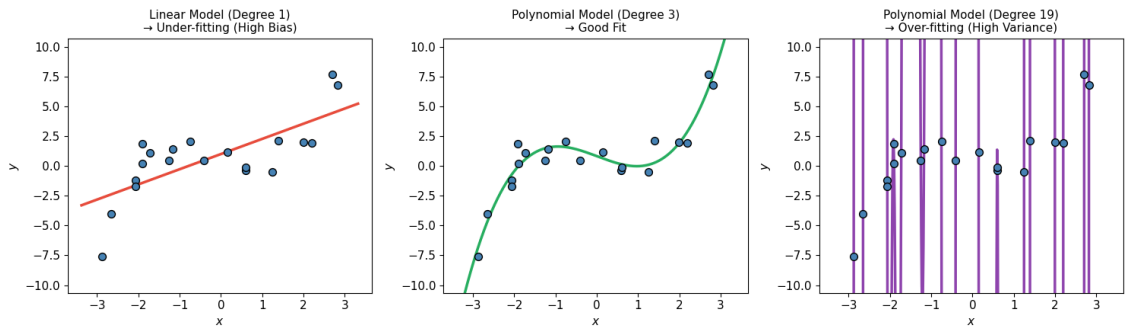
A critical aspect of machine learning is choosing the right model complexity. Too simple → underfitting. Too complex → overfitting. Let's visualize this with polynomial regression.

1.3.1 The Bias-Variance Trade-off

- **Underfitting (High Bias):** Model is too simple to capture patterns → poor performance on both training and test data
- **Overfitting (High Variance):** Model memorizes noise → excellent training performance, poor test performance
- **Sweet Spot:** Balanced complexity that generalizes well to new data

Below, we generate 20 noisy samples from a cubic polynomial and fit models of varying complexity:

[2]:
```
# Generate sample data from a cubic polynomial with noise
X, y = generate_polynomial_data(n=20, noise=1.0, seed=42)

# Visualize the three regimes: underfitting, good fit, overfitting
plot_model_selection_regression(X, y)
plt.tight_layout()
plt.show()

print(f"\n{sep}\nPolynomial Regression - Model Complexity:\n{sep}")
print("Left:   Linear model (underfitting) - captures the trend but misses␣
 ↪curvature")
print("Middle: Cubic model (good fit) - reasonable complexity")
print("Right:  Degree-19 polynomial (overfitting) - wiggles to fit every point")
```




```
<Figure size 1000x600 with 0 Axes>


==================================================
Polynomial Regression - Model Complexity:
==================================================
Left:   Linear model (underfitting) - captures the trend but misses curvature
Middle: Cubic model (good fit) - reasonable complexity
Right:  Degree-19 polynomial (overfitting) - wiggles to fit every point
```

1.3.2 Key Insight: Polynomial Features as Feature Engineering

Notice how the cubic model captures the underlying pattern well. This is an example of **feature engineering**—we created polynomial features (x^2, x^3) from the original feature x. This is our first hint that domain knowledge and feature creation can substantially improve model performance.

Counterinuitive take-away: Linear Regression can capture non-linear relationships if we engineer the right features. The

$ Polynomial Regression: y = _0 + _1 x + _2 x^2 + _3 x^3 + $

is linear in parameters, but non-linear in features!

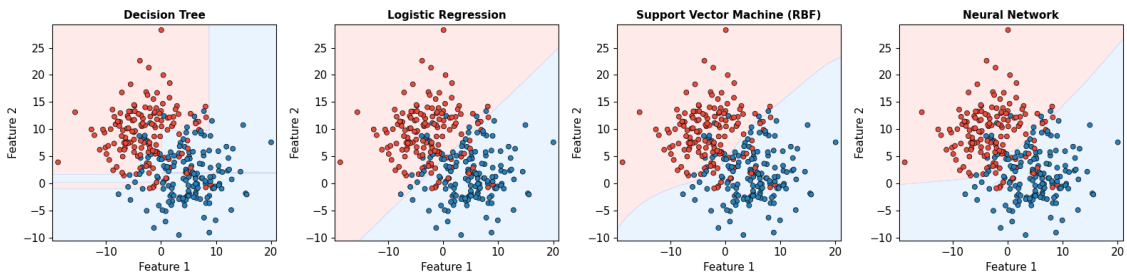
1.4 3. Model Selection — Classification (Decision Boundaries)

Different algorithms learn different decision boundaries. Let's explore how four popular classifiers separate two blob classes:

[3]:
```
# Generate two-blob classification data
X, y = generate_two_blobs(n=300, seed=42)

# Define classifiers
classifiers = [
    DecisionTreeClassifier(random_state=42, max_depth=4),
    LogisticRegression(random_state=42, max_iter=1000),
    SVC(kernel='rbf', random_state=42),
    MLPClassifier(hidden_layer_sizes=(20, 20), max_iter=1000, random_state=42)
]

titles = [
    'Decision Tree',
    'Logistic Regression',
    'Support Vector Machine (RBF)',
    'Neural Network'
]
```

Now let's visualize how each classifier draws its decision boundary on the same data.

[4]:
```
# Plot decision boundaries
plot_classification_boundaries(X, y, classifiers, titles, figsize=(16, 4))
plt.show()

print(f"\n{sep}\nDecision Boundaries:\n{sep}")
print("Decision Tree:        Axis-aligned splits, piecewise constant␣
 ↪boundaries")
print("Logistic Regression:  Linear boundary (can't capture non-linear␣
 ↪patterns)")
print("SVM (RBF):            Non-linear, smooth boundary")
print("Neural Network:       Flexible, complex boundary")
```



```
==================================================
Decision Boundaries:
==================================================
Decision Tree:        Axis-aligned splits, piecewise constant boundaries
Logistic Regression:  Linear boundary (can't capture non-linear patterns)
SVM (RBF):            Non-linear, smooth boundary
Neural Network:       Flexible, complex boundary
```

1.4.1 Observations

1. **Decision Tree** creates rectangular regions (axis-aligned splits)
2. **Logistic Regression** produces a linear boundary—works here, but not always
3. **Support Vector Machine** learns smooth, non-linear boundaries (RBF kernel)
4. **Neural Network** is highly flexible and can capture intricate patterns

The "best" algorithm depends on your data structure, interpretability needs, and computational constraints.

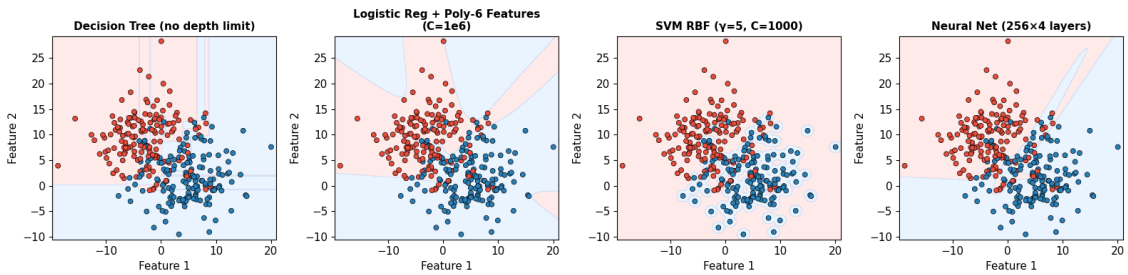
1.4.2 Same Four Classifiers — Now Deliberately Overfitted

What happens when we crank each model's complexity to the extreme?

Classifier	What we changed	Effect
Decision Tree	`max_depth=None` (unlimited)	Grows until every leaf is pure → super-jagged boundary
Logistic Regression	Degree-6 polynomial features + very weak regularisation (`C=1e6`)	Linear model in a 27-dim feature space → wiggly boundary
SVM (RBF)	`gamma=5, C=1000`	Each point gets a tiny "bubble" of influence → boundary hugs every point
Neural Network	4 hidden layers of 256 units, 5 000 iterations	Massive capacity → boundary contorts to fit noise

[5]:
```python
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import PolynomialFeatures

# Same data as before
X_blobs, y_blobs = generate_two_blobs(n=300, seed=42)

# Overfitted classifiers
classifiers_overfit = [
    DecisionTreeClassifier(random_state=42, max_depth=None),          # no␣
↪depth limit
    make_pipeline(
        PolynomialFeatures(degree=6, include_bias=False),
        LogisticRegression(C=1e6, max_iter=5000, random_state=42)     #␣
↪near-zero regularisation
    ),
    SVC(kernel='rbf', gamma=5, C=1000, random_state=42),             # tiny RBF␣
↪width, huge C
```

```
    MLPClassifier(hidden_layer_sizes=(256, 256, 256, 256),
                  max_iter=5000, random_state=42)                    # very␣
 ↪large network
]

titles_overfit = [
    'Decision Tree (no depth limit)',
    'Logistic Reg + Poly-6 Features\n(C=1e6)',
    'SVM RBF ( =5, C=1000)',
    'Neural Net (256×4 layers)'
]

plot_classification_boundaries(X_blobs, y_blobs, classifiers_overfit,␣
 ↪titles_overfit, figsize=(16, 4))
plt.suptitle('Overfitted Decision Boundaries', fontsize=14, fontweight='bold',␣
 ↪y=1.02)
plt.show()

print(f"\n{sep}\nOverfitted Decision Boundaries:\n{sep}")
print("All four models now hug the training data far too closely.")
print("Decision Tree:      Extremely jagged, axis-aligned rectangles␣
 ↪everywhere")
print("Logistic Reg+Poly:  Wiggly boundary from high-dim feature space")
print("SVM (RBF):          Tight 'bubbles' around individual points")
print("Neural Network:     Complex, contorted boundary capturing noise")
```




```
<Figure size 1000x600 with 0 Axes>


==================================================
Overfitted Decision Boundaries:
==================================================
All four models now hug the training data far too closely.
Decision Tree:      Extremely jagged, axis-aligned rectangles everywhere
Logistic Reg+Poly:  Wiggly boundary from high-dim feature space
SVM (RBF):          Tight 'bubbles' around individual points
Neural Network:     Complex, contorted boundary capturing noise
```

1.5 4. Data Encoding

Most machine learning algorithms require numerical inputs. Categorical variables must be encoded:

Feature Type	Encoding Method	Example
Nominal (unordered)	One-Hot Encoding	Color: Red → [1,0,0], Blue → [0,1,0]
Ordinal (ordered)	Label/Map	Risk level: Low→1, Medium→2, High→3
Numerical	No encoding	Age, Claim amount

1.5.1 One-Hot Encoding (Nominal Features)

For unordered categories, create a binary column per category:

```
import pandas as pd
from sklearn.preprocessing import OneHotEncoder

df = pd.DataFrame({'policy_type': ['car', 'home', 'car', 'life']})
df_encoded = pd.get_dummies(df, columns=['policy_type'])
```

1.5.2 Ordinal Encoding (Ordered Features)

For ordered categories, map values to integers that preserve the order:

```
df = pd.DataFrame({'risk_level': ['low', 'medium', 'high', 'low']})
risk_map = {'low': 1, 'medium': 2, 'high': 3}
df['risk_level_encoded'] = df['risk_level'].map(risk_map)
```

1.5.3 *Reading Exercise 1: Data Encoding*

Task: Read, how a small DataFrame containing categorical features is encoded.

[6]:
```
# Sample insurance data
df_insurance = pd.DataFrame({
    'policy_type': ['car', 'home', 'car', 'life', 'home'],
    'risk_level': ['low', 'high', 'medium', 'low', 'high'],
    'age': [25, 45, 35, 60, 50]
})

print(f"\n{sep}\nOriginal DataFrame:\n{sep}")
print(df_insurance)


# One-hot encode 'policy_type' using pd.get_dummies
df_encoded = pd.get_dummies(df_insurance, columns=['policy_type'],␣
 ↪drop_first=False)

# Ordinal encode 'risk_level' using .map()
risk_mapping = {'low': 1, 'medium': 2, 'high': 3}
df_encoded['risk_level_encoded'] = df_encoded['risk_level'].map(risk_mapping)
df_encoded = df_encoded.drop('risk_level', axis=1)

print(f"\n{sep}\nEncoded DataFrame:\n{sep}")
print(df_encoded)
```

```
==================================================
Original DataFrame:
==================================================
  policy_type risk_level  age
0         car        low   25
1        home       high   45
2         car     medium   35
3        life        low   60
4        home       high   50

==================================================
Encoded DataFrame:
==================================================
   age  policy_type_car  policy_type_home  policy_type_life  risk_level_encoded
0   25             True             False             False                   1
1   45            False              True             False                   3
2   35             True             False             False                   2
3   60            False             False              True                   1
4   50            False              True             False                   3
```

1.6 5. Scaling / Standardization

Many algorithms (e.g., SVM, Neural Networks, Logistic Regression) are sensitive to feature magnitude. Scaling brings features to comparable ranges.

1.6.1 StandardScaler (Standardization)

$$z = \frac{x - \mu}{\sigma}$$

Transforms features to mean=0, standard deviation=1.

1.6.2 MinMaxScaler (Normalization)

$$x' = \frac{x - x_{\text{min}}}{x_{\text{max}} - x_{\text{min}}}$$

Scales features to the range [0, 1].

1.6.3 Code Example

[7]:
```
from sklearn.preprocessing import StandardScaler, MinMaxScaler

# Sample data
X_sample = np.array([[10, 1000],
                     [20, 2000],
                     [30, 3000]])

print(f"\n{sep}\nOriginal Data:\n{sep}")
print(X_sample)
```

```
==================================================
Original Data:
==================================================
[[  10 1000]
 [  20 2000]
 [  30 3000]]
```

StandardScaler transforms each feature to have mean 0 and standard deviation 1.

[8]:
```python
# Standardization
scaler_standard = StandardScaler()
X_standardized = scaler_standard.fit_transform(X_sample)
print(f"\n{sep}\nAfter StandardScaler (mean 0, std 1):\n{sep}")
print(X_standardized)
```

```
==================================================
After StandardScaler (mean 0, std 1):
==================================================
[[-1.22474487 -1.22474487]
 [ 0.          0.        ]
 [ 1.22474487  1.22474487]]
```

MinMaxScaler rescales each feature to the range [0, 1].

[9]:
```python
# Normalization
scaler_minmax = MinMaxScaler()
X_normalized = scaler_minmax.fit_transform(X_sample)
print(f"\n{sep}\nAfter MinMaxScaler (range [0,1]):\n{sep}")
print(X_normalized)
```

```
==================================================
After MinMaxScaler (range [0,1]):
==================================================
[[0.  0. ]
 [0.5 0.5]
 [1.  1. ]]
```

1.7 6. Feature Engineering

Feature engineering is the art of creating new features from raw data. It often has bigger impact on model performance than algorithm choice.

1.7.1 We've Already Seen Feature Engineering!

In Section 2, we created polynomial features (x, x^2, x^3) from a single input. This transformation allowed a cubic model to fit the data.

1.7.2 Actuarial Examples

- **Combined Features:** `claim_ratio = claim_amount / vehicle_value`
- **Polynomial Features:** `age^2` to capture non-linear age effects

- **Interactions:** `premium_group * vehicle_type` to capture combined effects of nominal variables as matrices (e.g. young drivers in sports cars)
- **Domain Ratios:** `average_claim_per_policy = total_claims / policy_count`

1.7.3 Key Principle

Domain knowledge is your best tool. Understand your data, consult with subject matter experts, and craft features that capture meaningful patterns.

1.8 7. Train/Test Split

A fundamental principle: evaluate on data the model has **never seen** before.

1.8.1 The Workflow

1. Split data: typically 80% training, 20% testing
2. Fit the model on training data
3. Evaluate on test data

1.8.2 Critical: Avoid Data Leakage

Data Leakage occurs when test set information "leaks" into training:

```
#  WRONG: Scaling before split
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)  # Fit on ALL data
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2)

#  CORRECT: Scaling after split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)      # Fit only on training
X_test_scaled = scaler.transform(X_test)            # Transform test using training statistic
```

[10]:
```
# Example: correct train/test split and scaling
from sklearn.datasets import load_iris

iris = load_iris()
X_iris = iris.data
y_iris = iris.target

# Step 1: Split first
X_train, X_test, y_train, y_test = train_test_split(
    X_iris, y_iris, test_size=0.2, random_state=42
)

print(f"\n{sep}\nTrain/Test Split:\n{sep}")
print(f"Training set size: {X_train.shape[0]}")
print(f"Test set size:     {X_test.shape[0]}")

# Step 2: Scale using only training statistics
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
print(f"\n{sep}\nResult:\n{sep}")
print("  Data correctly split and scaled (no leakage)")
```

```
==================================================
Train/Test Split:
==================================================
Training set size: 120
Test set size:     30

==================================================
Result:
==================================================
  Data correctly split and scaled (no leakage)
```

1.9 8. Evaluation Metrics

The choice of metric depends on your problem and business objectives.

1.9.1 Regression: Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

1.9.2 Classification Metrics

For a binary classification problem (e.g., claim/no-claim):

1.9.3 Confusion Matrix

	Predicted Positive	**Predicted Negative**
Actual Positive	TP (True Positive)	FN (False Negative)
Actual Negative	FP (False Positive)	TN (True Negative)

1.9.4 Derived Metrics

- **Accuracy:** $\frac{\text{TP+TN}}{\text{Total}}$ — Overall correctness
- **Precision:** $\frac{\text{TP}}{\text{TP+FP}}$ — Of predicted positives, how many are correct?
- **Recall:** $\frac{\text{TP}}{\text{TP+FN}}$ — Of actual positives, how many did we find?
- **F1-Score:** $2 \cdot \frac{\text{Precision}\cdot\text{Recall}}{\text{Precision+Recall}}$ — Harmonic mean

1.9.5 When to Use Each Metric

Metric	**When to Use**
Accuracy	Balanced classes, general purpose
Precision	Minimize false positives (e.g., emergency brake system)
Recall	Minimize false negatives (e.g., disease diagnosis)
F1-Score	Need a balance between precision and recall

Let's see these metrics in action:

[11]:
```
from sklearn.metrics import classification_report, accuracy_score

# Example predictions vs. actual labels
y_true = np.array([1, 0, 1, 1, 0, 1, 1, 0, 1, 1])
y_pred = np.array([0, 0, 1, 0, 0, 1, 1, 1, 1, 1])

print(f"\n{sep}\nClassification Report:\n{sep}")
print(classification_report(y_true, y_pred, target_names=['No Claim (0)',␣
↪'Claim (1)']))
```

```
==================================================
Classification Report:
==================================================
              precision    recall  f1-score   support

No Claim (0)       0.50      0.67      0.57         3
   Claim (1)       0.83      0.71      0.77         7

    accuracy                           0.70        10
   macro avg       0.67      0.69      0.67        10
weighted avg       0.73      0.70      0.71        10
```

We can also compute individual metrics for more targeted analysis.

[12]:
```
from sklearn.metrics import precision_score, recall_score, f1_score

# Individual metrics
acc = accuracy_score(y_true, y_pred)
precision = precision_score(y_true, y_pred)
recall = recall_score(y_true, y_pred)
f1 = f1_score(y_true, y_pred)

print(f"\n{sep}\nIndividual Metrics:\n{sep}")
print(f"Accuracy:  {acc:.3f}")
print(f"Precision: {precision:.3f}")
print(f"Recall:    {recall:.3f}")
print(f"F1-Score:  {f1:.3f}")
```

```
==================================================
Individual Metrics:
==================================================
Accuracy:  0.700
Precision: 0.833
Recall:    0.714
F1-Score:  0.769
```

1.9.6 Nerd Talk: Why Do the Two Classes Get Different Scores?

Precision and recall differ between "No Claim (0)" and "Claim (1)" because the errors are **not symmetric**. It helps to swap the perspective:

- **FP for Claim (1) = FN for No Claim (0)** — predicting "Claim (1)" when there is none hurts *precision* of "Claim (1)" but hurts *recall* of "No Claim (0)".
- **FN for Claim (1) = FP for No Claim (0)** — missing a real claim hurts *recall* of "Claim (1)" but hurts *precision* of "No Claim (0)".

Every false positive for one class is a false negative for the other. So the same confusion matrix, read row-by-row, yields a different precision/recall pair for each class.

Viewed from "Claim (1)" (positive = 1):

	Pred Claim (1)	**Pred No Claim (0)**
Actual Claim (1)	TP = 5	FN = 2
Actual No Claim (0)	FP = 1	TN = 2

→ Precision = 5/(5+1) = 0.83 · Recall = 5/(5+2) = 0.71

Viewed from "No Claim (0)" (positive = 0) — just swap the labels:

	Pred No Claim (0)	**Pred Claim (1)**
Actual No Claim (0)	TP = 2	FN = 1
Actual Claim (1)	FP = 2	TN = 5

→ Precision = 2/(2+2) = 0.50 · Recall = 2/(2+1) = 0.67

Same numbers, different names — that's why each class gets its own precision and recall.

1.10 9. Decision Trees with scikit-learn

Decision Trees are interpretable, easy to use, and effective across many domains.

1.10.1 The sklearn Interface

All sklearn estimators follow a consistent interface:

```
clf = SomeClassifier()           # Create instance
clf.fit(X_train, y_train)         # Fit to training data
y_pred = clf.predict(X_test)      # Make predictions
score = clf.score(X_test, y_test) # Evaluate
```

1.10.2 *Reading Exercise 3: Train and Evaluate a Decision Tree*

Task: Train a Decision Tree classifier on the blob data, split it properly, and generate a classification report.

[13]:
```
# Use the two-blob data from earlier
X, y = generate_two_blobs(n=300, seed=42)

# Step 1: Split the data
```

```python
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Step 2: Create and train a Decision Tree
tree_clf = DecisionTreeClassifier(max_depth=5, random_state=42)
tree_clf.fit(X_train, y_train)

# Step 3: Make predictions
y_pred = tree_clf.predict(X_test)

# Step 4: Evaluate
print(f"\n{sep}\nDecision Tree - Classification Report:\n{sep}")
print(classification_report(y_test, y_pred))

print(f"\n{sep}\nTest Accuracy:\n{sep}")
print(f"{tree_clf.score(X_test, y_test):.3f}")
```

```
==================================================
Decision Tree - Classification Report:
==================================================
              precision    recall  f1-score   support

           0       0.68      0.72      0.70        29
           1       0.72      0.68      0.70        31

    accuracy                           0.70        60
   macro avg       0.70      0.70      0.70        60
weighted avg       0.70      0.70      0.70        60


==================================================
Test Accuracy:
==================================================
0.700
```

1.11 10. Random Forests

A **Random Forest** is an ensemble of many decision trees. Instead of relying on a single tree, we:

1. Build multiple trees on bootstrap samples of the data
2. Introduce randomness (e.g., random feature subsets for splits)
3. Average predictions across all trees

1.11.1 Key Advantages

- **Reduced Overfitting:** Averaging multiple trees is more robust than a single tree
- **Black-box Model:** Often performs very well, but less interpretable
- **Parallelizable:** Trees are independent and can be trained in parallel

1.11.2 Comparison: Single Tree vs. Random Forest

[14]:
```
# Use the blob data - this time with more noise
X, y = generate_two_blobs(n=300, seed=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Single Decision Tree
tree_clf = DecisionTreeClassifier(max_depth=5, random_state=42)
tree_clf.fit(X_train, y_train)
tree_acc = tree_clf.score(X_test, y_test)
```

Now let's train a Random Forest — an ensemble of many decision trees — on the same data and compare.

[15]:
```
# Random Forest
rf_clf = RandomForestClassifier(n_estimators=5, max_depth=5, random_state=42)
rf_clf.fit(X_train, y_train)
rf_acc = rf_clf.score(X_test, y_test)

print(f"\n{sep}\nDecision Tree vs. Random Forest:\n{sep}")
print(f"Decision Tree Test Accuracy:  {tree_acc:.3f}")
print(f"Random Forest Test Accuracy:  {rf_acc:.3f}")

print(f"\n{sep}\nRandom Forest Benefits:\n{sep}")
print("- Often achieves better accuracy")
print("- More robust to overfitting")
print("- Can handle large feature sets")
print("- Slower to train than single tree, but still fast for most datasets")
```

```
==================================================
Decision Tree vs. Random Forest:
==================================================
Decision Tree Test Accuracy:  0.700
Random Forest Test Accuracy:  0.817

==================================================
Random Forest Benefits:
==================================================
- Often achieves better accuracy
- More robust to overfitting
- Can handle large feature sets
- Slower to train than single tree, but still fast for most datasets
```

1.12 11. Neural Networks (MLP)

A **Multi-Layer Perceptron (MLP)** is a simple feedforward neural network. It consists of:

1. An **input layer** that receives the features
2. One or more **hidden layers** of neurons, each applying a non-linear activation function
3. An **output layer** that produces the prediction

1.12.1 How It Works

Each neuron computes a weighted sum of its inputs, adds a bias, and passes the result through an activation function (typically ReLU). During training, the network adjusts all weights via **back-propagation** to minimize the loss.

1.12.2 Key Characteristics

Aspect	MLP
Strengths	Flexible, can learn complex non-linear patterns
Weaknesses	Requires scaled data, less interpretable, sensitive to hyperparameters
When to use	When simpler models underperform and you have enough data

In scikit-learn, `MLPClassifier` provides a simple interface — no deep learning framework needed.

1.12.3 Comparison: Decision Tree vs. Random Forest vs. Neural Network

[16]:
```
# Reuse the same blob data and split from Section 10
# X_train, X_test, y_train, y_test are already defined

# The MLP needs scaled data (just like Logistic Regression and SVM)
scaler_mlp = StandardScaler()
X_train_scaled_mlp = scaler_mlp.fit_transform(X_train)
X_test_scaled_mlp = scaler_mlp.transform(X_test)

# Neural Network: one hidden layer with 16 neurons, early stopping to prevent␣
↪overfitting
mlp_clf = MLPClassifier(
    hidden_layer_sizes=(16,),      # single hidden layer, 16 neurons
    activation='relu',             # ReLU activation function
    solver='adam',                 # Adam optimizer
    max_iter=500,                  # maximum training epochs
    early_stopping=True,           # stop when validation score stops improving
    random_state=42
)
mlp_clf.fit(X_train_scaled_mlp, y_train)
mlp_acc = mlp_clf.score(X_test_scaled_mlp, y_test)

# Compare all three models
print(f"\n{sep}\nModel Comparison on Blob Data:\n{sep}")
print(f"  Decision Tree:     {tree_acc:.3f}")
print(f"  Random Forest:     {rf_acc:.3f}")
print(f"  Neural Network:    {mlp_acc:.3f}")

print(f"\n{sep}\nMLP Details:\n{sep}")
print(f"Architecture:    {mlp_clf.hidden_layer_sizes} hidden layer(s)")
print(f"Stopped at epoch: {mlp_clf.n_iter_} / {mlp_clf.max_iter}")

print(f"\n{sep}\nNote:\n{sep}")
```

```python
print("On this simple 2D dataset, all three models perform")
print("similarly well. The MLP's advantage becomes more apparent")
print("with higher-dimensional data and complex decision boundaries.")
```

```
==================================================
Model Comparison on Blob Data:
==================================================
  Decision Tree:     0.700
  Random Forest:     0.817
  Neural Network:    0.850

==================================================
MLP Details:
==================================================
Architecture:    (16,) hidden layer(s)
Stopped at epoch: 33 / 500

==================================================
Note:
==================================================
On this simple 2D dataset, all three models perform
similarly well. The MLP's advantage becomes more apparent
with higher-dimensional data and complex decision boundaries.
```

1.13 12. Classifiers in scikit-learn (selection)

1.13.1 Quick Reference: Classifiers in scikit-learn

We've now seen Decision Trees, Random Forests, and Neural Networks in action. Here is a broader overview of classifiers available in scikit-learn — all follow the same `.fit()` / `.predict()` interface:

Algorithm	sklearn class	Key characteristics
Logistic Regression	`LogisticRegression`	Linear, fast, interpretable coefficients, good baseline
Decision Tree	`DecisionTreeClassifier`	Non-linear, interpretable rules, prone to overfitting
Random Forest	`RandomForestClassifier`	Ensemble of trees, robust, handles many features well
Gradient Boosting	`GradientBoostingClassifier`	Sequential ensemble, often top accuracy, slower to train
AdaBoost	`AdaBoostClassifier`	Boosting with weak learners, sensitive to outliers
SVM (linear)	`LinearSVC`	Linear boundary, fast on large datasets, no probabilities
SVM (kernel)	`SVC`	Non-linear via kernels (RBF, poly), needs scaling
k-Nearest Neighbors	`KNeighborsClassifier`	Instance-based, no training phase, slow at prediction time
Naive Bayes	`GaussianNB`	Very fast, assumes feature independence, good for text

Algorithm	sklearn class	Key characteristics
Neural Network	`MLPClassifier`	Flexible non-linear, needs scaling, many hyperparameters

Rule of thumb: Start with Logistic Regression (interpretable baseline), then try Random Forest or Gradient Boosting. Use SVMs or MLPs if you need non-linear boundaries and can afford less interpretability.

1.14 13. Hyperparameter Optimization with GridSearchCV

Hyperparameters are settings you choose before training (e.g., `max_depth`, `n_estimators`). Finding good values is crucial.

GridSearchCV systematically tries combinations and uses cross-validation to estimate performance.

1.14.1 Example: Optimizing Random Forest

[17]:
```
from sklearn.model_selection import GridSearchCV

# Use the blob data
X, y = generate_two_blobs(n=300, seed=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Define a grid of hyperparameters to search
param_grid = {
    'max_depth': [3, 5, 7],
    'n_estimators': [3, 15, 50]
}

# Create base classifier
rf = RandomForestClassifier(random_state=42)

# GridSearchCV tries all combinations
grid_search = GridSearchCV(rf, param_grid, cv=5, scoring='accuracy', n_jobs=-1)
grid_search.fit(X_train, y_train)

print(f"\n{sep}\nGridSearchCV Results:\n{sep}")
print(f"Best parameters:          {grid_search.best_params_}")
print(f"Best cross-validation score: {grid_search.best_score_:.3f}")
print(f"Test set score (best params): {grid_search.score(X_test, y_test):.3f}")
```

```
==================================================
GridSearchCV Results:
==================================================
Best parameters:          {'max_depth': 3, 'n_estimators': 3}
Best cross-validation score: 0.846
Test set score (best params): 0.833
```

1.14.2 What's Happening?

1. GridSearchCV creates a $4 \times 3 = 12$ combinations of hyperparameters
2. For each, it uses 5-fold cross-validation on the training set
3. It selects the combination with the best average CV score
4. The best estimator is retrained on all training data and returned
5. We evaluate the final model on the held-out test set

Tip: Use GridSearchCV inside your train/test pipeline to avoid data leakage.

1.15 14. Feature Importance — Which Features Actually Matter?

After training a model, a natural question arises: **which features drive the predictions?**

This is especially important for actuaries, where: - **Regulators** may require explanations for pricing or reserving decisions - **Business stakeholders** want to understand the key risk drivers - **Model validation** needs to confirm that important features make domain sense

1.15.1 Permutation Importance (Model-Agnostic)

Permutation Importance measures how much a model's performance drops when a single feature is randomly shuffled. The idea is simple:

1. Compute the model's baseline score on the test set
2. For each feature: shuffle its values, re-score the model, and measure the drop
3. A large drop means the feature is important — the model relied on it

This approach works with **any** model (not just tree-based ones) and evaluates importance on held-out data, making it more trustworthy than training-based measures.

We use `sklearn.inspection.permutation_importance` to compute this.

1.15.2 Example: Titanic Survival Prediction

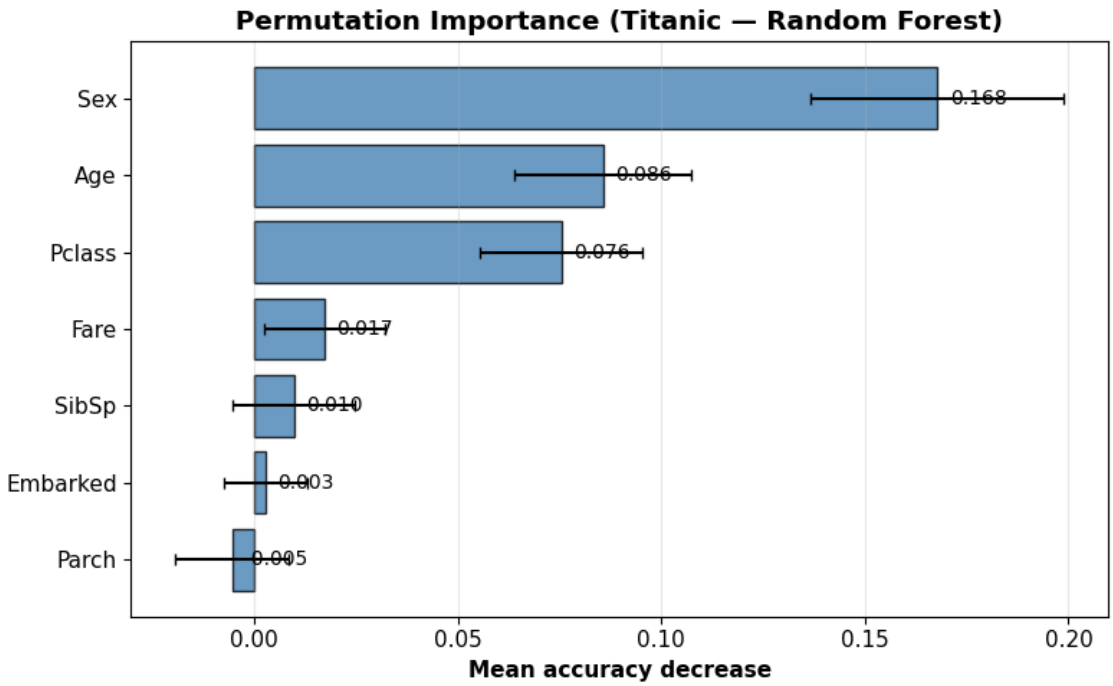
The Titanic dataset is a classic — who survived the sinking? Features include passenger class, sex, age, fare, and family size. Let's find out which ones the model actually relies on.

[18]:
```
from sklearn.inspection import permutation_importance

# --- Load and prepare Titanic data ---
X_titanic, y_titanic, feature_names = load_titanic_data()

print(f"\n{sep}\nTitanic Dataset:\n{sep}")
print(f"Passengers: {len(y_titanic)}")
print(f"Features:   {feature_names}")
print(f"Survival rate: {y_titanic.mean():.1%}")

# Split
X_train_t, X_test_t, y_train_t, y_test_t = train_test_split(
    X_titanic, y_titanic, test_size=0.2, random_state=42
)

# Train a Random Forest
rf_titanic = RandomForestClassifier(n_estimators=100, random_state=42)
rf_titanic.fit(X_train_t, y_train_t)
```

```python
baseline_acc = rf_titanic.score(X_test_t, y_test_t)

print(f"\n{sep}\nBaseline Test Accuracy:\n{sep}")
print(f"{baseline_acc:.3f}")

# Compute Permutation Importance on the TEST set
perm_result = permutation_importance(
    rf_titanic, X_test_t, y_test_t,
    n_repeats=30, random_state=42, scoring='accuracy'
)

# Sort by importance
sorted_idx = perm_result.importances_mean.argsort()[::-1]

# Plot
fig, ax = plt.subplots(figsize=(8, 5))
y_pos = np.arange(len(feature_names))
ax.barh(y_pos, perm_result.importances_mean[sorted_idx],
        xerr=perm_result.importances_std[sorted_idx],
        color='steelblue', alpha=0.8, edgecolor='black', capsize=3)
ax.set_yticks(y_pos)
ax.set_yticklabels([feature_names[i] for i in sorted_idx])
ax.invert_yaxis()
ax.set_xlabel('Mean accuracy decrease', fontweight='bold')
ax.set_title('Permutation Importance (Titanic - Random Forest)',␣
 ↪fontweight='bold')
ax.grid(axis='x', alpha=0.3)

# Add value labels
for i, idx in enumerate(sorted_idx):
    val = perm_result.importances_mean[idx]
    ax.text(val + 0.003, i, f'{val:.3f}', va='center', fontsize=10)

plt.tight_layout()
plt.show()

# Print results
print(f"\n{sep}\nPermutation Importance (mean ± std over 30 repeats):\n{sep}")
for i in sorted_idx:
    mean = perm_result.importances_mean[i]
    std = perm_result.importances_std[i]
    print(f"  {feature_names[i]:12s}  {mean:.4f} ± {std:.4f}")
```

```
==================================================
Titanic Dataset:
==================================================
Passengers: 712
Features:   ['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']
Survival rate: 40.4%
```

```
==================================================
Baseline Test Accuracy:
==================================================
0.783
```




```
==================================================
Permutation Importance (mean ± std over 30 repeats):
==================================================
  Sex           0.1678 ± 0.0310
  Age           0.0858 ± 0.0217
  Pclass        0.0755 ± 0.0199
  Fare          0.0172 ± 0.0147
  SibSp         0.0098 ± 0.0150
  Embarked      0.0028 ± 0.0101
  Parch         -0.0054 ± 0.0139
```

1.15.3 Interpreting the Results

Sex is by far the strongest predictor — "women and children first" was a real policy on the Titanic. **Fare** and **Pclass** (passenger class) also rank high, reflecting the well-documented survival advantage of wealthier passengers with cabins on upper decks. **Age** contributes moderately, consistent with the priority given to children. Features like **Embarked** and **SibSp** have minimal impact.

1.15.4 Key Takeaways

- **Permutation Importance is model-agnostic:** It works with any classifier or regressor — not just tree-based models. You can apply the same technique to Logistic Regression, SVMs, or Neural Networks.

- **Always compute on the test set:** Using the training set can give misleading results for overfit models.
- **Error bars matter:** The `n_repeats` parameter (here: 30) gives us standard deviations. A feature with high mean importance but also high variance deserves closer investigation.
- **Watch out for correlated features:** If two features carry similar information, permutation importance may underestimate both — because shuffling one still leaves the other intact. In actuarial practice, this is common (e.g., age and years of driving experience). On the Titanic, Fare and Pclass are correlated, which may explain why neither alone shows extreme importance.
- **Actuarial relevance:** Feature importance helps explain model decisions to regulators, validates that the model captures genuine risk drivers, and highlights which data is most valuable to collect and maintain.

1.16 15. *Reading Exercise: Comprehensive Classification Pipeline*

Task: Build classification reports for multiple algorithms on the blob dataset.

You will see: 1. Generate two-blob data 2. Split into train/test 3. Train Logistic Regression, SVM, and Neural Network 4. Evaluate each with classification_report 5. Compare performance

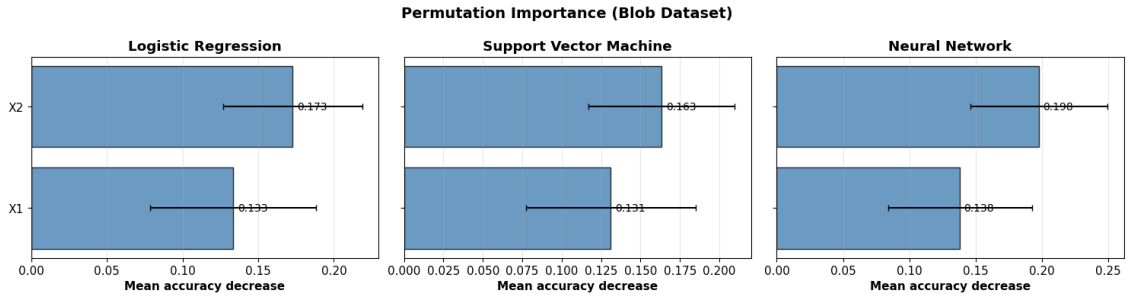
[19]:
```
# Step 1: Generate and split data
X, y = generate_two_blobs(n=300, seed=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
↪random_state=42)

# Step 2: Scale the data (important for LR, SVM, NN)
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Step 3: Define classifiers with hyperparameter grids for tuning
param_grids = {
    'Logistic Regression': {
        'C': [0.01, 0.1, 1, 10],
        'solver': ['lbfgs']
    },
    'Support Vector Machine': {
        'C': [0.1, 1, 10],
        'gamma': ['scale', 0.1, 1]
    },
    'Neural Network': {
        'hidden_layer_sizes': [(10,), (20, 20), (50, 20)],
        'alpha': [0.0001, 0.001, 0.01]
    }
}

base_classifiers = {
    'Logistic Regression': LogisticRegression(max_iter=1000, random_state=42),
    'Support Vector Machine': SVC(kernel='rbf', random_state=42),
    'Neural Network': MLPClassifier(max_iter=1000, random_state=42)
}
```

```python
# Step 4: Hyperparameter tuning with GridSearchCV, then evaluate
blob_feature_names = ['X1', 'X2']
classifiers = {}   # will hold the best estimators
results = {}

for name in base_classifiers:
    grid_search = GridSearchCV(
        base_classifiers[name], param_grids[name],
        cv=5, scoring='accuracy', n_jobs=-1
    )
    grid_search.fit(X_train_scaled, y_train)

    # Store the best estimator
    best_clf = grid_search.best_estimator_
    classifiers[name] = best_clf

    # Evaluate on the test set
    y_pred = best_clf.predict(X_test_scaled)
    acc = accuracy_score(y_test, y_pred)
    results[name] = acc

    print(f"\n{sep}\n{name}:\n{sep}")
    print(f"Best parameters:          {grid_search.best_params_}")
    print(f"Best CV score:            {grid_search.best_score_:.3f}")
    print(f"Test Accuracy:            {acc:.3f}")
    print(f"\n{sep}\nClassification Report:\n{sep}")
    print(classification_report(y_test, y_pred, target_names=['Class 0', 'Class␣
↪1']))

print(f"\n{sep}\nSummary of Accuracies (after GridSearchCV tuning):\n{sep}")
for name, acc in results.items():
    print(f"  {name:30s}: {acc:.3f}")

# Step 5: Feature Importance (Permutation Importance for each tuned classifier)
from sklearn.inspection import permutation_importance

fig, axes = plt.subplots(1, len(classifiers), figsize=(5 * len(classifiers),␣
↪4), sharey=True)

for ax, (name, clf) in zip(axes, classifiers.items()):
    perm_result = permutation_importance(
        clf, X_test_scaled, y_test,
        n_repeats=30, random_state=42, scoring='accuracy'
    )
    sorted_idx = perm_result.importances_mean.argsort()[::-1]
    y_pos = np.arange(len(blob_feature_names))

    ax.barh(y_pos, perm_result.importances_mean[sorted_idx],
            xerr=perm_result.importances_std[sorted_idx],
            color='steelblue', alpha=0.8, edgecolor='black', capsize=3)
    ax.set_yticks(y_pos)
```

```python
    ax.set_yticklabels([blob_feature_names[i] for i in sorted_idx])
    ax.invert_yaxis()
    ax.set_xlabel('Mean accuracy decrease', fontweight='bold')
    ax.set_title(name, fontweight='bold')
    ax.grid(axis='x', alpha=0.3)

    for i, idx in enumerate(sorted_idx):
        val = perm_result.importances_mean[idx]
        ax.text(val + 0.003, i, f'{val:.3f}', va='center', fontsize=10)

plt.suptitle('Permutation Importance (Blob Dataset)', fontweight='bold',␣
 ↪fontsize=14)
plt.tight_layout()
plt.show()

# Print permutation importance for each classifier
for name, clf in classifiers.items():
    perm_result = permutation_importance(
        clf, X_test_scaled, y_test,
        n_repeats=30, random_state=42, scoring='accuracy'
    )
    sorted_idx = perm_result.importances_mean.argsort()[::-1]
    print(f"\n{sep}\nPermutation Importance - {name} (mean ± std over 30␣
 ↪repeats):\n{sep}")
    for i in sorted_idx:
        mean = perm_result.importances_mean[i]
        std = perm_result.importances_std[i]
        print(f"  {blob_feature_names[i]:12s}  {mean:.4f} ± {std:.4f}")
```

```
==================================================
Logistic Regression:
==================================================
Best parameters:          {'C': 0.1, 'solver': 'lbfgs'}
Best CV score:            0.846
Test Accuracy:            0.833

==================================================
Classification Report:
==================================================
              precision    recall  f1-score   support

     Class 0       0.85      0.79      0.82        29
     Class 1       0.82      0.87      0.84        31

    accuracy                           0.83        60
   macro avg       0.84      0.83      0.83        60
weighted avg       0.83      0.83      0.83        60


==================================================
```

```
Support Vector Machine:
==================================================
Best parameters:          {'C': 0.1, 'gamma': 'scale'}
Best CV score:            0.858
Test Accuracy:            0.817

==================================================
Classification Report:
==================================================
              precision    recall  f1-score   support

     Class 0       0.88      0.72      0.79        29
     Class 1       0.78      0.90      0.84        31

    accuracy                           0.82        60
   macro avg       0.83      0.81      0.81        60
weighted avg       0.82      0.82      0.81        60


==================================================
Neural Network:
==================================================
Best parameters:          {'alpha': 0.0001, 'hidden_layer_sizes': (50, 20)}
Best CV score:            0.854
Test Accuracy:            0.833

==================================================
Classification Report:
==================================================
              precision    recall  f1-score   support

     Class 0       0.91      0.72      0.81        29
     Class 1       0.78      0.94      0.85        31

    accuracy                           0.83        60
   macro avg       0.85      0.83      0.83        60
weighted avg       0.85      0.83      0.83        60


==================================================
Summary of Accuracies (after GridSearchCV tuning):
==================================================
  Logistic Regression           : 0.833
  Support Vector Machine        : 0.817
  Neural Network                : 0.833
```




```
==================================================
Permutation Importance - Logistic Regression (mean ± std over 30 repeats):
==================================================
  X2            0.1728 ± 0.0458
  X1            0.1333 ± 0.0549

==================================================
Permutation Importance - Support Vector Machine (mean ± std over 30 repeats):
==================================================
  X2            0.1633 ± 0.0464
  X1            0.1311 ± 0.0539

==================================================
Permutation Importance - Neural Network (mean ± std over 30 repeats):
==================================================
  X2            0.1978 ± 0.0518
  X1            0.1383 ± 0.0543
```

1.16.1 Reflection

- Which classifier performed best? Why might that be?
- Did scaling help? (Try commenting out the scaling step for SVM and NN—what happens?)
- What trade-offs exist between accuracy and interpretability?
- In an actuarial context, which metric (accuracy, precision, recall) might matter most?

1.17 Summary

In this 90-minute session, you have learned:

1. **Learning Paradigms:** Supervised (classification & regression), Unsupervised, Reinforcement
2. **Model Complexity:** Balance between underfitting (bias) and overfitting (variance)
3. **Data Preparation:** Encoding, scaling, feature engineering
4. **Workflow Best Practices:** Train/test split, avoiding data leakage
5. **Evaluation:** Metrics for regression and classification
6. **Algorithms:** Decision Trees, Random Forests, Logistic Regression, SVM, Neural Networks
7. **Optimization:** Using GridSearchCV to tune hyperparameters
8. **Feature Importance:** Using Permutation Importance to understand which features drive predictions

1.17.1 Key Takeaways for Actuaries

- Machine learning can enhance reserve prediction, pricing, and risk assessment
- Careful data preparation (encoding, scaling, feature engineering) often matters more than algorithm choice
- Always evaluate on held-out test data
- **Feature importance** helps explain model decisions to regulators and validates that models capture genuine risk drivers
- Interpretability is often valued in actuarial applications—consider decision trees and logistic regression
- Domain expertise drives feature engineering—collaborate with your risk team

1.17.2 Next Steps

- Explore real actuarial datasets
- Experiment with different feature engineering strategies
- Use permutation importance to validate your models and communicate results
- Consider your specific business objectives when choosing evaluation metrics
- Stay curious and iterate!

Created for: Introduction to Machine Learning for Actuaries
Duration: 90 minutes
Tools: Python, scikit-learn, pandas, matplotlib